

Polargraph Chalk Plotter

Cable-driven mechatronics for large-format wall drawing

MECHATRONICS	MOTION CONTROL	FABRICATION
--------------	----------------	-------------

Project overview

A cable-driven drawing machine that converts planar artwork into coordinated motor motion and deposits chalk on a vertical surface. The project combines inverse kinematics, step generation, end-effector design, calibration, and structural iteration.

Engineering problem

Large-format plotting is mechanically simple in concept but sensitive to cable stretch, spool geometry, wall friction, missed steps, and chalk contact force.

System architecture - Two upper motor-spool assemblies control cable lengths to position the carriage. - A chalk carriage provides compliance, repeatable contact, and a controlled lift mechanism. - A motion planner converts vector paths into short line segments and synchronized steps. - A calibration routine maps commanded cable length to measured wall coordinates.	Engineering focus - Inverse kinematics from target (x, y) position to left and right cable lengths. - Step-density and acceleration limits selected to reduce oscillation and missed steps. - Spool-radius compensation and homing references for repeatability.
--	---

Verification plan

- Plot a calibration grid and measure scale, skew, closure, and repeatability errors.
- Test circles, diagonals, corners, and long strokes at increasing speed.
- Log failure modes such as cable slip, chalk chatter, and carriage tilt.

Status and evidence boundary Summer 2026 mechatronics prototype brief; quantitative accuracy belongs in a later test report.
--